

# **Retail Sales Intelligence & Profitability Analysis Dashboard**

## **Professional Project Report**

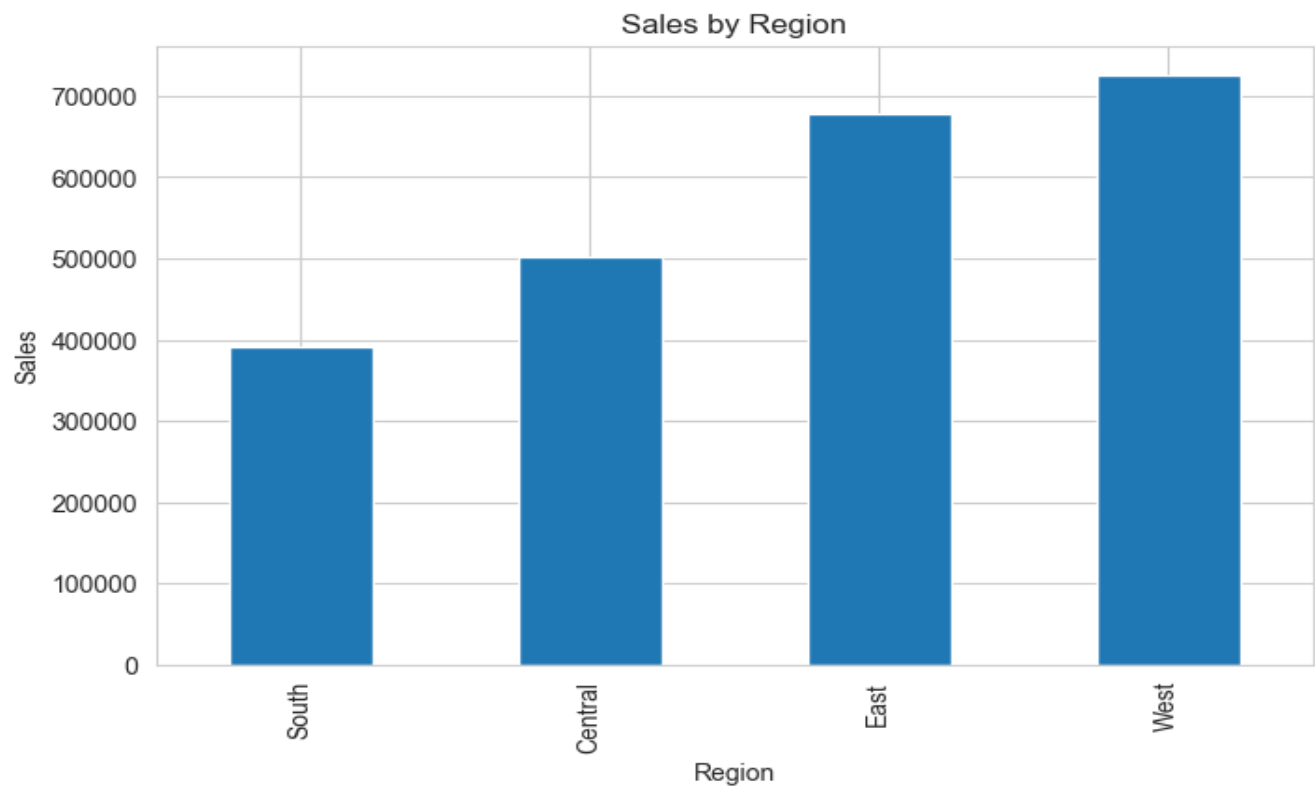
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**Data Analytics Internship Project**

## **Executive Summary**

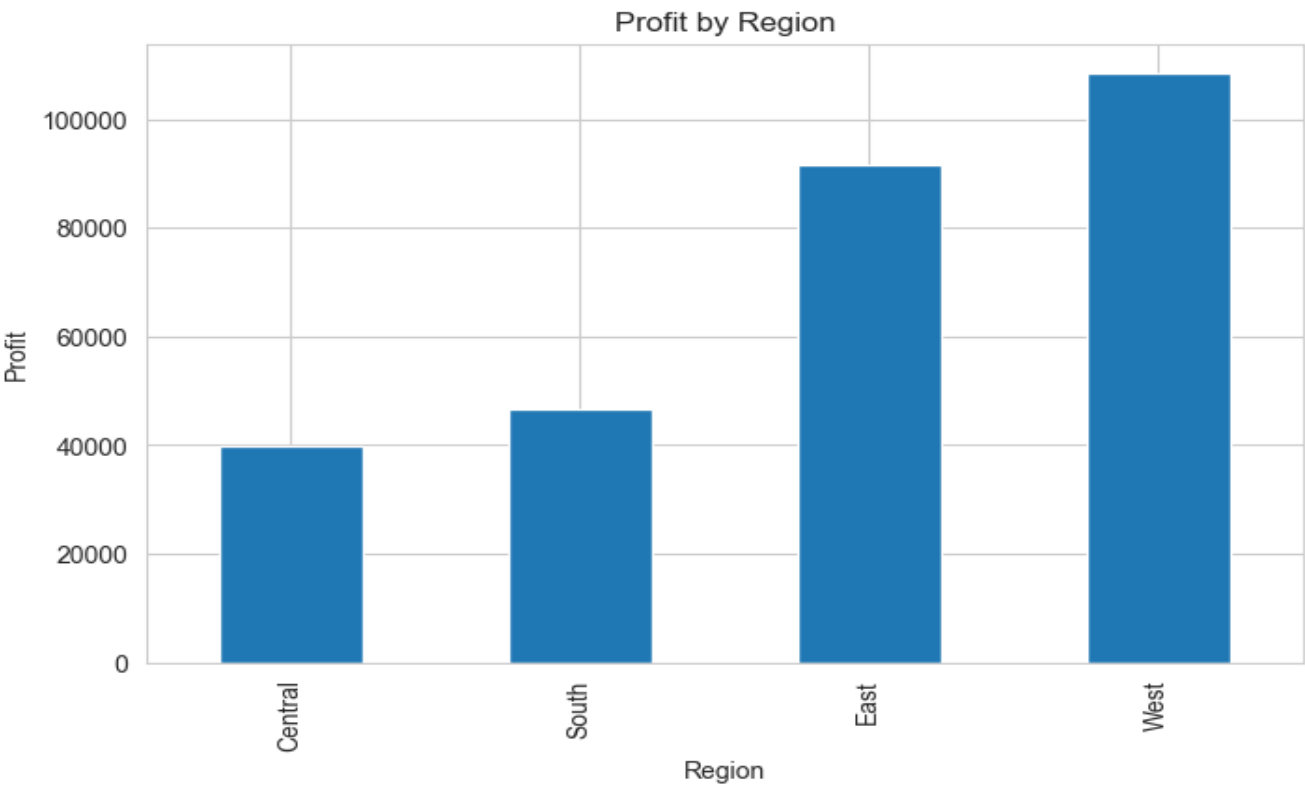
This project analyzes retail sales performance, regional profitability, category contribution, discount impact, and business opportunities using exploratory data analysis and business intelligence techniques. The objective is to convert raw retail data into actionable insights that support strategic decision-making and profitability improvement.

# Sales by Region



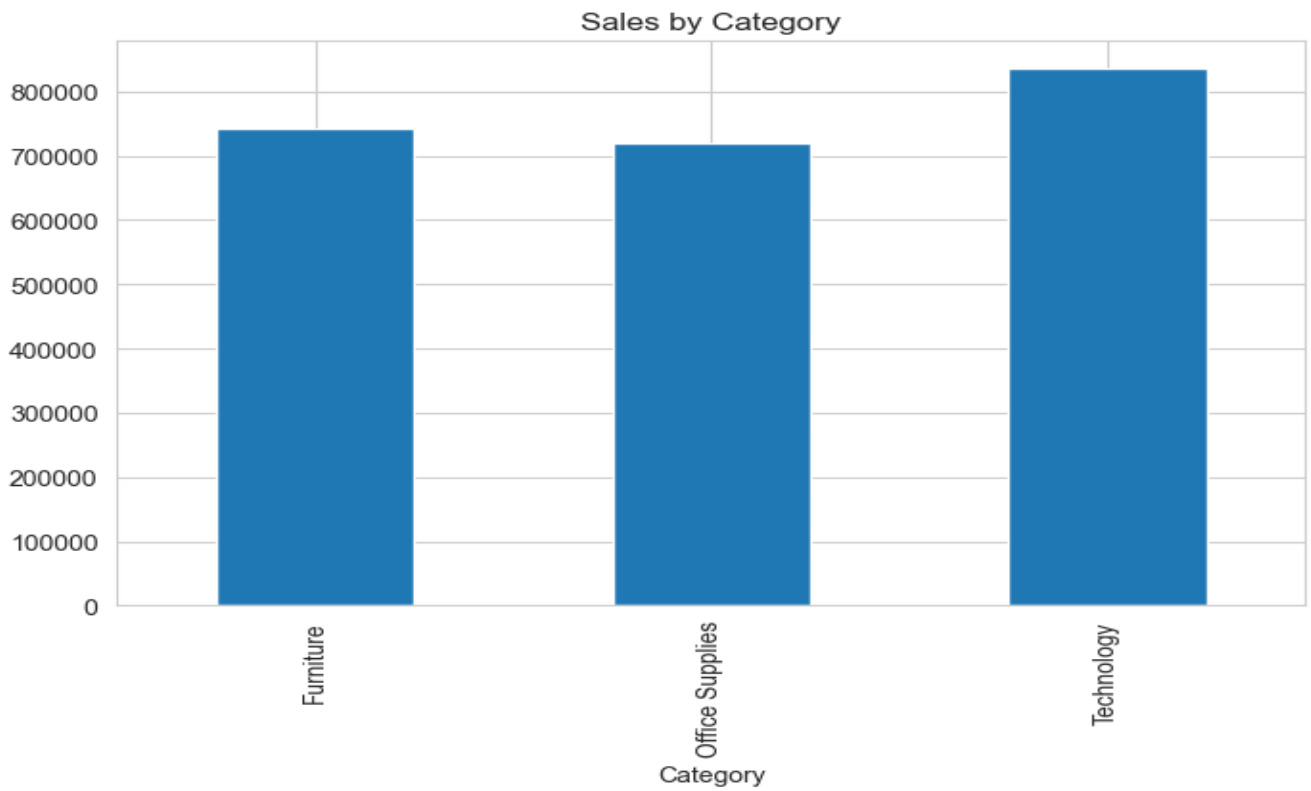
Regional sales analysis shows the distribution of revenue across different markets. The West region emerged as the strongest contributor, indicating high customer demand and strong business performance.

# Profit by Region



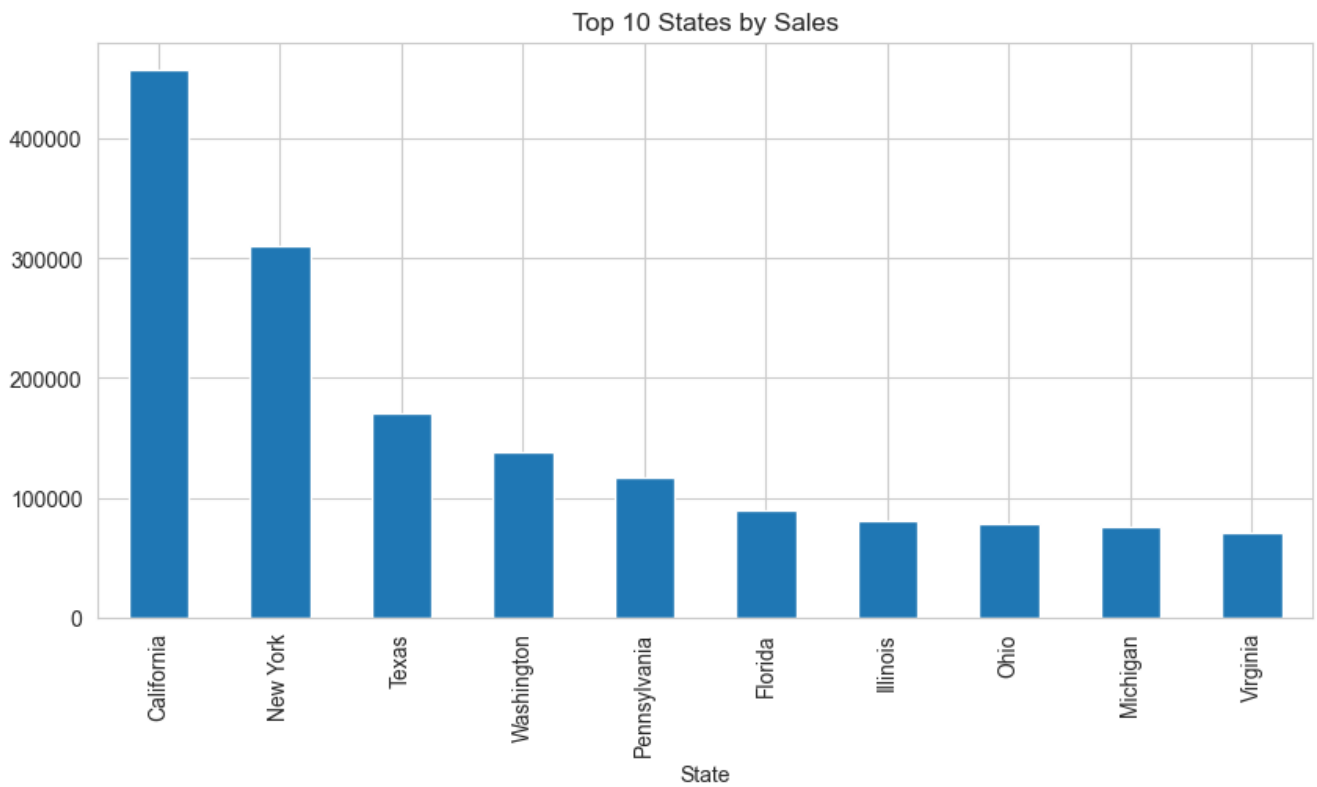
Profitability analysis identifies regions generating the highest earnings. Strong-performing regions represent potential areas for expansion and investment.

# Sales by Category



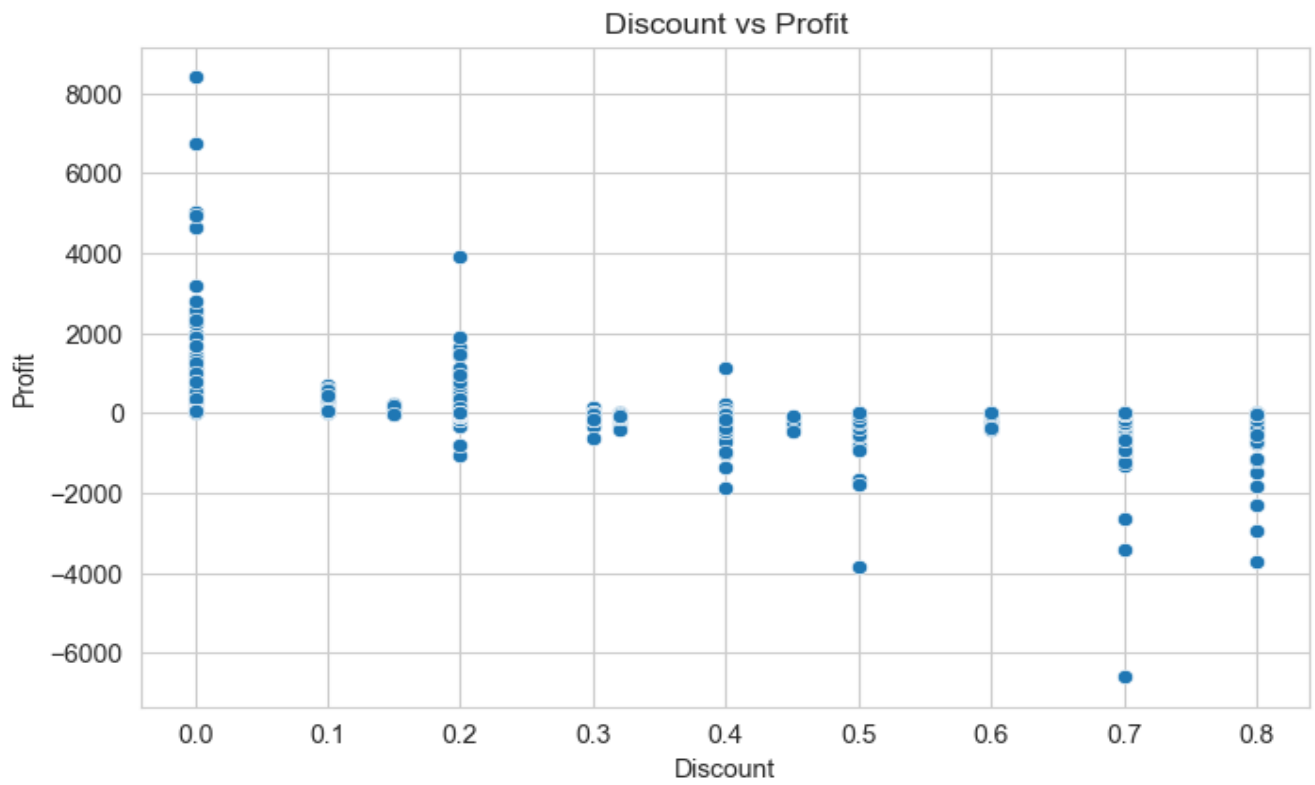
Category analysis highlights which product groups contribute most to total sales and helps prioritize inventory and marketing strategies.

# Top States by Sales



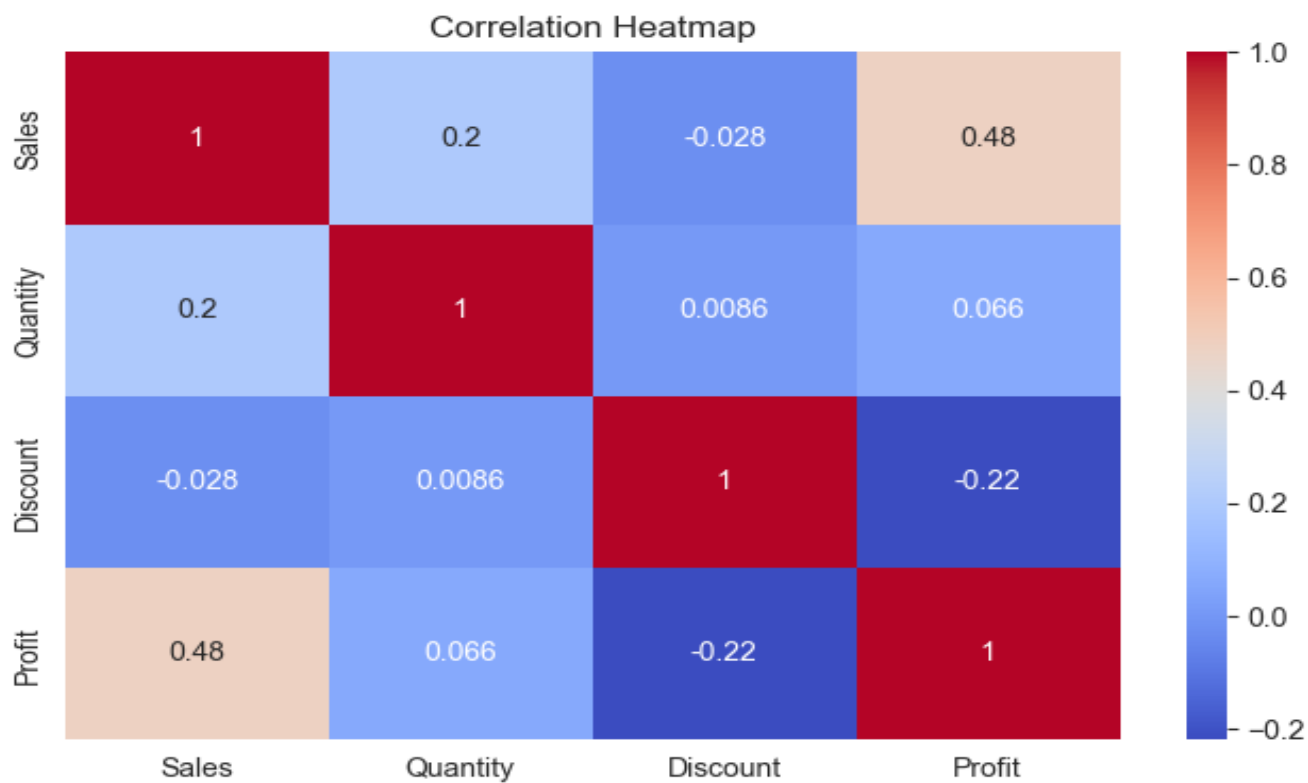
State-level analysis reveals the most valuable markets and supports location-based business planning.

# Discount vs Profit



The relationship between discount and profit demonstrates how excessive discounting can reduce profitability despite generating sales.

# Correlation Heatmap





## Key Findings

- West region generated the highest sales and profit.
- East region emerged as the second strongest market.
- Discounts negatively affected profitability.
- Sales and Profit showed a positive relationship.
- High-performing states contributed a significant share of overall revenue.
- Category performance varied across the business.

## **Business Recommendations**

1. Increase investment in high-performing regions.
2. Optimize discount strategies to improve margins.
3. Focus marketing efforts on profitable categories.
4. Monitor underperforming markets and categories.
5. Use analytics-driven decision making for future growth.

## **Conclusion**

The project demonstrates how retail data can be transformed into valuable business intelligence. Through regional, category, profitability, and correlation analysis, important trends were identified that can support strategic planning, revenue growth, and operational optimization.